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A CASE SERIES OF GRANULOMATOUS MASTITIS OF THE BREAST: A RADIOLOGIC PERSPECTIVE ON A DIAGNOSTIC CHALLENGE

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BACKGROUND

Granulomatous mastitis (GM) is an uncommon inflammatory breast disease that poses a significant diagnostic challenge due to its ability to closely mimic breast malignancy. The incidence of GM appears to be increasing, although its exact causes remain unknown. According to Shabani S et al., autoimmune or hypersensitivity reactions are the likely hypotheses. From January to December 2025, our centre diagnosed a total of 41 biopsy-proven cases of GM, from which we retrospectively reviewed the clinical records, multimodality imaging findings, and histopathological outcomes of four selected female patients aged 33 to 63 years.

CASE PRESENTATION

Each case presented with a palpable breast lump but demonstrated a distinct sonographic pattern. Case 1 presented as a highly suspicious, irregular lobulated hypoechoic mass with focal asymmetry and architectural distortion on mammography (BI-RADS 4c). Case 2 presented as a persistent heterogeneous hypoechoic collection (BI-RADS 4a). Case 3 presented as an oval, circumscribed complex cyst with a soft-tissue component (BI-RADS 4a), while Case 4 presented as a complicated cyst with mobile internal echogenicity (BI-RADS 3). Case 4 underwent biopsy due to patient concern.

DISCUSSION

GM was first described by Kessler and Wolloch in 1972, with a reported incidence of approximately 2.4%—occurring more frequently in Asian, Turkish, Hispanic, and East Asian populations.

Imaging features of GM vary significantly. Mammographic features range from a completely normal appearance to non-specific findings, including mass-like lesions, focal asymmetric densities, or architectural distortion. Ultrasound demonstrates a wider range of findings, including abscess formation, mass-like or tubular/nodular hypoechogenicity, focal parenchymal heterogeneity, and hypervascularity on Color Doppler. MRI findings, such as mass or non-mass enhancement, rim-enhancing abscess-like lesions and restricted diffusion, are similarly non-specific.

The differential diagnosis for GM includes infectious etiology (e.g., Mycobacterium tuberculosis, fungal, bacterial, or parasitic infections) and non-infectious causes (idiopathic, foreign-body reaction, trauma, autoimmune, or systemic disease). Tropical climate and hygiene conditions are hypothesised as contributing factors.

In Malaysia, particularly in Sabah where tuberculosis remains endemic, distinguishing GM from tuberculous mastitis (TM) is critical because management strategies differ fundamentally: GM is typically managed with observation, corticosteroid therapy, immunosuppressive therapy, or surgery, whereas TM requires antituberculous therapy.

In our series, histopathological examination confirmed granulomatous inflammation across all cases, notably with preserved normal ipsilateral axillary lymph nodes. The primary histological distinction lies in the presence of non-caseating granulomas in GM versus caseating granulomas in TM.

FIGURES

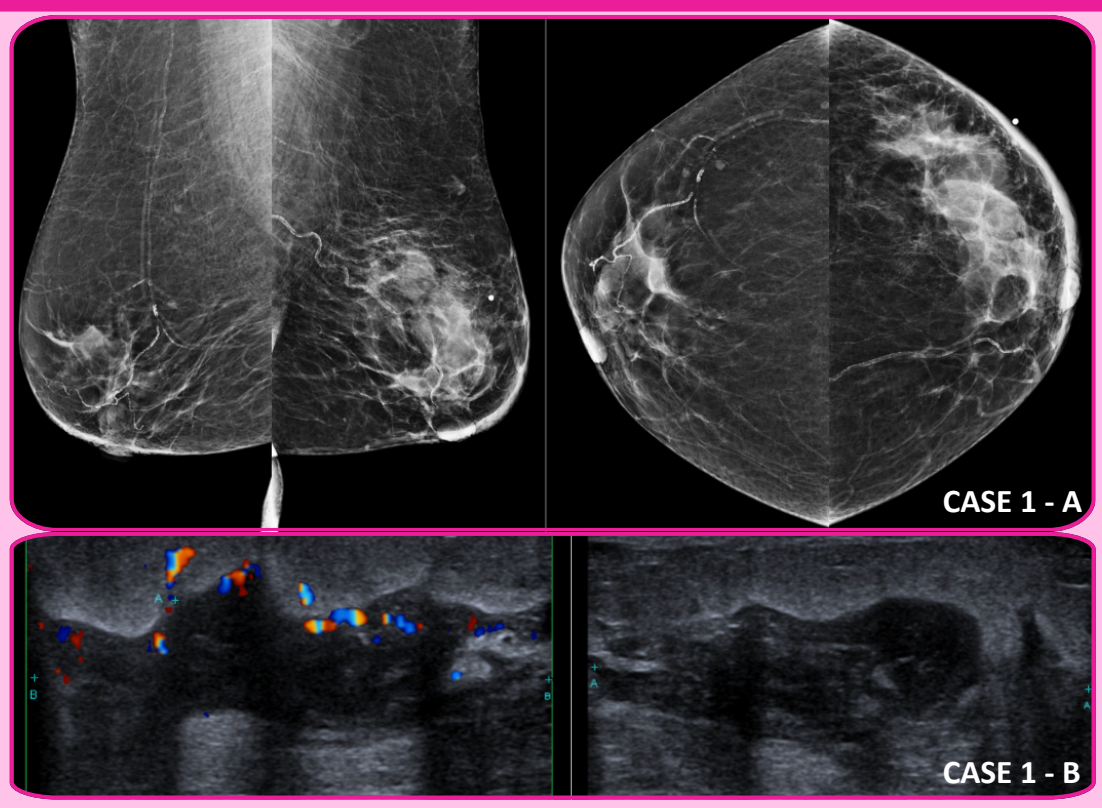

Figure 1:
(A) Mammogram images showing heterogeneously dense breast parenchymal with focal asymmetry associated with distortion and diffuse skin thickening in the left outer breast, anterior to middle third (BB marker region). No dominant mass, concerning calcification or distortion in the right breast.
(B) Ultrasound images showing irregular, lobulated hypoechoic mass with minimal intralesional vascularity in the left upper outer breast, 5.5cm in widest diameter.

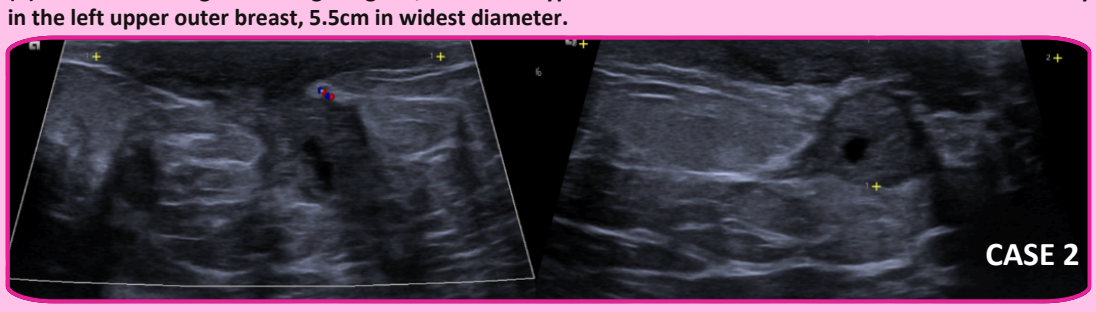

Figure 2: Ultrasound images showing heterogeneous hypoechoic collection with tubular hypoechogenicity and minimal perilesional vascularity seen along the left 9 o'clock radian, 5.8cm in widest diameter.

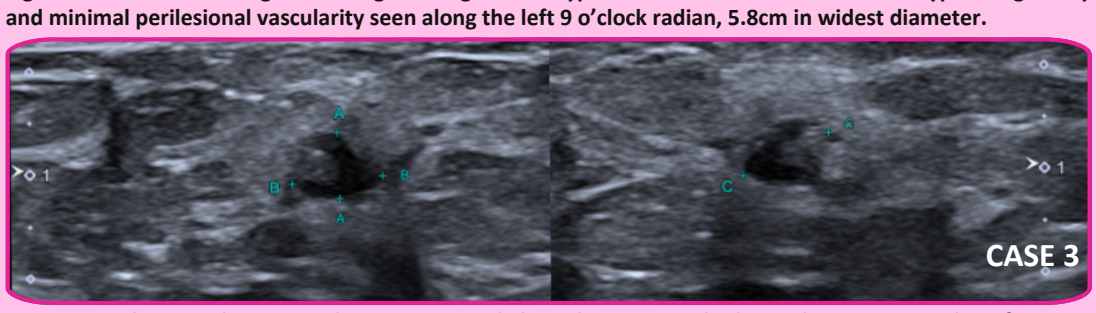

Figure 3: Ultrasound images showing an oval-shaped, circumscribed anechoic mass with soft tissue component within along the right 7 o'clock radian, 0.5cm in widest diameter.

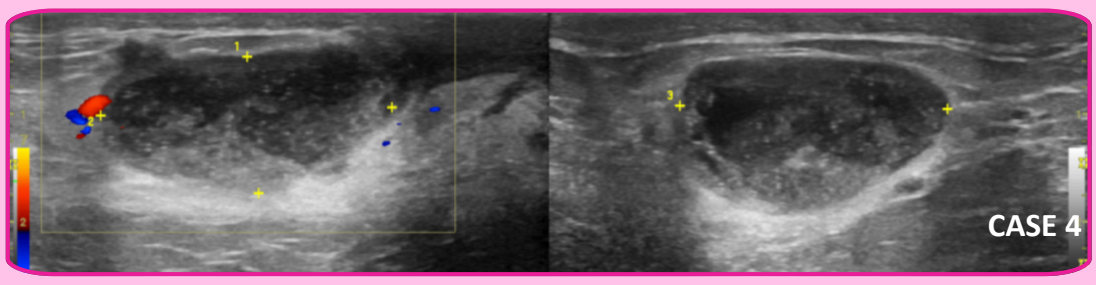

Figure 4: Ultrasound images showing an oval-shaped, circumscribed anechoic mass with mobile internal echogenicity within along the left 9 o'clock radian, 2.7cm in widest diameter.

CONCLUSION

As GM displays diverse imaging features ranging from simple fluid collections to highly suspicious masses without pathognomonic radiological signs, it poses a diagnostic challenge, particularly in a tuberculosis-endemic region such as Sabah. Early recognition and meticulous histopathological evaluation are essential to distinguish idiopathic granulomatous mastitis from tuberculous mastitis, avoiding misdiagnosis and ensuring appropriate patient management.

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